电源提供能量 $\int_0^\infty U_s i\,dt = U_s q = CU_s^2$

电阻消耗能量 $\int_0^\infty i^2 R\,dt = \int_0^\infty \left(\frac{U_s}{R} e^{-\frac{t}{RC}}\right)^2 R\,dt$

$$= \int_0^\infty \frac{U_s^2}{R^2} e^{-\frac{2t}{RC}} \cdot R\,dt = \frac{U_s^2}{R}\int_0^\infty e^{-\frac{2t}{RC}}\,dt$$

$$= \frac{U_s^2}{R}\left[-\frac{RC}{2}\cdot e^{-\frac{2t}{RC}}\Big|_0^\infty\right] = \frac{U_s^2}{R}\left[-\frac{RC}{2}\cdot 0 + \frac{RC}{2}\cdot 1\right]$$

$$= \frac{1}{2}CU_s^2$$

电容储存能量 $\frac{1}{2}CU_s^2$